

Expectation requires treatment to boost pain relief: An fMRI study



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ABSTRACT

We investigated the effect of a possible interaction between topical analgesic treatment and treatment expectation on pain at the behavioral and neuronal level by combining topical lidocaine/prilocaine treatment with an expectancy manipulation in a 2 by 2 within-subject design (open treatment, hidden treatment, placebo, control). Thirty-two healthy subjects received heat pain stimuli on capsaicin-pretreated skin and rated their experienced pain during functional magnetic resonance imaging. This allowed us to separate drug- and expectancy-related effects at the behavioral and neuronal levels and to test whether they interact during the processing of painful stimuli. Pain ratings were reduced during active treatment and were associated with reduced activity in the anterior insular cortex. Pain ratings were lower in open treatment compared with hidden treatment and were related to reduced activity in the anterior insular cortex, the anterior cingulate cortex, the secondary somatosensory cortex, and the thalamus. Testing for an interaction revealed that the expectation effect was significantly larger in the active treatment conditions compared with the no-treatment conditions and was associated with signal changes in the anterior insular cortex, the anterior cingulate cortex, and the ventral striatum. In conclusion, this study shows that even in the case of a topical analgesic, expectation interacts with treatment at the level of pain ratings and neuronal responses in placebo-related brain regions. Our results are highly relevant in the clinical context as they show (i) that expectation can boost treatment and (ii) that expectation and treatment are not necessarily additive as assumed in placebo-controlled clinical trials.

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1. Introduction

Therapeutic outcomes such as pain relief after administration of an analgesic drug do not merely reflect pure pharmacological effects – they also depend on contextual factors, such as expectation, hope or beliefs about the treatment [7,19,40]. The relationship between specific drug and contextual factors is of high theoretical and clinical interest. From a therapeutic point of view, it is desirable to maximize treatment effects and so, apart from selecting the appropriate treatment, it is important to exploit positive contextual contributions to treatment outcome as much as possible [16,39]. From a methodological perspective, the interplay of pharmacological and contextual effects is crucial, as current randomized clinical trials are based on the assumption of the additivity of pharmacological and contextual effects [5]. However, the precise interplay of pharmacotherapy and cognitive factors is still insufficiently understood. Although drug and expectancy interactions

have been observed in different fields [27,29,34,54], few studies have so far examined the precise interplay of analgesic drug effects with expectations.

A recent study has shown that positive and negative expectancies are able to modulate the analgesic treatment outcome during an infusion of the μ -opioid-receptor agonist remifentanyl [9]. However, this study could not formally establish an interaction, as the treatment (ie, drug) was not varied. Another recent study used different remifentanyl levels and found evidence for an additive effect, but no interaction between drug and expectation, on pain ratings and brain responses [4]. Both studies investigated opiate treatment. Given that that expectation and experience of pain relief is related to the release of endogenous opioids, an additive effect can be expected as the endogenous and exogenous opioids might simply add together [1,18,31]. Nevertheless, synergistic effects could be present with regard to a different substance/receptor system [6] or a specific psychological mediator (eg, conditioning), or owing to a positive feedback between drug effects and psychological factors (eg, reinforcement of expectations through the awareness of incipient pain relief). Considering the latter point, drugs would not necessarily have to enter the central nervous system to generate an interaction effect.

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The aim of the present study was to investigate a possible interaction between local analgesic treatment and expectation effects on pain perception and associated neuronal processes. We used heat pain stimuli on capsaicin-sensitized skin, an established model of clinical pain [20,49], and combined topical lidocaine/prilocaine treatment (T) with treatment expectancy manipulations (E) in a balanced placebo design [32,42]. This leads to 4 conditions: open treatment (T+E+), hidden treatment (T+E−), placebo (T−E+), and control (T−E−). We recorded pain ratings, expectancy ratings, and BOLD responses using functional magnetic resonance imaging (fMRI). This allowed us to separate drug-related and expectancy-related effects at the behavioral and neuronal level and to test whether they interact during the processing of painful stimuli.

2. Materials and methods

2.1. Subjects

Thirty-nine young, healthy subjects participated in this study. Seven subjects did not complete data collection owing to technical difficulties or an abnormal heat pain threshold (on capsaicin). All analyses were performed on the remaining 32 subjects (mean age 25.6 years $\pm$ 3.3, range 19–33, 17 male, 15 female). Exclusion criteria were neurological diseases (including pain syndromes), skin afflictions on the forearms, current medication (excluding oral contraceptives), substance abuse, and pregnancy. None of the subjects reported relevant depressive symptoms (Beck Depression Inventory II > 14). The study was approved by the Ethics Committee of the Medical Council of Hamburg and all participants gave written consent in accordance with the Declaration of Helsinki.

2.2. Experimental paradigm

Subjects were recruited using an Internet advertisement platform frequently visited by students of the University of Hamburg. Subjects were told that this study investigated the influence of an analgesic cream (“lidocaine/prilocaine, a very effective pain killer”) on brain responses evoked by painful thermal stimulation on capsaicin-sensitized skin as a model of clinical pain. An initial telephone screening was used to determine eligibility for the experiment.

The study design combined an established placebo cream paradigm with lidocaine/prilocaine treatment to capsaicin-sensitized skin as an efficacious therapeutic model [36,50]. This allowed us to realize a within-subject 2 by 2 factorial design with the factors active treatment (T) and treatment expectancy (E). Accordingly, our study comprised 4 experimental conditions (Fig. 1A): open treatment (T+E+), hidden treatment (T+E−), placebo (T−E+), and control (T−E−).

First, subjects were informed about the experimental procedures and signed consent forms. Next, visual analog scale (VAS, described later) rating training was performed. Subsequently, 5 squares of 3 $\times$ 3 cm were marked on their left and right volar forearms in condition-corresponding colors (2 green for treatment expectancy, 2 red for no-treatment expectancy, 1 black for pain threshold) (Fig. 1A). For each condition, a separate skin patch was used (2 on the left, 2 on the right forearm).

Subjects were then visited by the experimenter, wearing a white coat, introducing lidocaine/prilocaine ointment as a highly effective analgesic treatment. Directly afterwards, subjects filled out a questionnaire concerning their expectancy regarding their anticipated pain with and without the announced treatment (0–10, end points labeled with “no pain at all” and “unbearable pain”), as well as questionnaires concerning their mood and anxiety level [26,51]. Ten minutes before entering the MRI scanner,

subjects received capsaicin cream (ABC-Wärmecreme, 0.075% capsaicin, Beiersdorf AG, Hamburg, Germany) on all 5 marked skin areas. Directly before entering the scanner, the residual capsaicin cream was removed from the skin.

Subsequently, the local analgesic cream (Emla, lidocaine/prilocaine ointment, AstraZeneca, Wedel, Germany) and an identical-looking control cream consisting of the same basic components without the active agents were applied. The application site and the order of conditions were counterbalanced across subjects in 4 pseudorandomized sequences. The study was conducted in a double-blind manner. Neither the subjects nor the experimenter knew the application sites of the active treatment. All skin patches were then covered with plasters.

Heat pain stimuli were applied using a Peltier thermode (TSAIL, Medoc, Ramat Yishai, Israel). Temperature calibration was performed to elicit a pain level of approximately 6 on a VAS (0–10). The calibration was performed on an additional skin patch on the left volar forearm on the basis of a heat pain threshold assessment. The heat pain threshold was assessed using the method of limits [21] with a slope of 0.3°C/s. Subjects had to press a button once they started to experience pain. Five additional blocks of 3 pain stimuli each (10 seconds duration) were used to approximate an individual stimulation temperature corresponding to VAS 6 on the previously mentioned scale for each subject. The range of the stimulation temperatures was 36–48°C. The same temperature was used for the individual subject during the whole experiment.

During the calibration procedure, subjects were lying in the scanner but no imaging measurements were performed. This allowed the subjects to adapt to the scanner environment and pain stimuli. Once the calibration was finished, spare cream was removed and all marked skin areas were again covered with plasters until needed for testing.

Control of the experimental timing and the stimulus presentation was achieved using Cogent 2000 v1.25 (Wellcome Department of Imaging Neuroscience, London, UK). Subjects operated a standard button box in order to provide responses. Behavioral data were collected using Matlab 2009B (Mathworks, Natick, MA, USA).

The fMRI measurements consisted of 4 sessions, each corresponding to one experimental condition. Each session consisted of 12 heat pain trials (see Fig. 1B). Each trial started with a condition cue (3 seconds), which, depending on the experimental condition, indicated whether the subject should expect treatment (E+) or not (E−) in that session. After a fixation period (white cross, 2 seconds), subjects evaluated their treatment expectancy on a VAS (0–10, end points labeled with “no pain relief” and “complete pain relief”, 10 seconds). This was followed by a fixation period (2 seconds). During the following pain anticipation phase, the fixation cross turned red, indicating that the painful stimulus would soon follow. The pain stimulus lasted for 20 seconds (1.5 seconds ramp up, 17 seconds plateau, 1.5 seconds ramp down). After a 5-second delay, subjects rated their perceived pain intensity on a VAS scale (0–10, end points labeled with “no pain at all” and “unbearable pain”, 10 seconds). The inter-trial interval varied between 10 and 15 seconds. Between sessions, the thermode was repositioned to the site corresponding to the subsequent experimental condition. After all 4 sessions had been completed, a T1-weighted anatomical scan was performed. The experiment was concluded with a debriefing outside the scanner. Subjects were informed about the preceding deception and were again asked for permission to use the collected data. None of the subjects withheld their approval.

2.3. MRI data acquisition

Imaging data were acquired on a 3 Tesla system (Magnetom TIM Trio, Siemens, Erlangen, Germany) equipped with a 32-channel head coil. To measure BOLD responses, a T2*-weighted

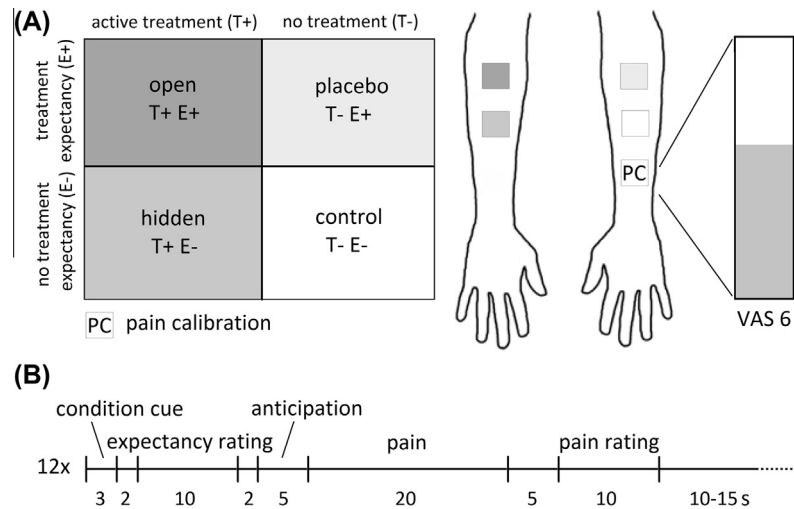


Fig. 1. (A) Within-subject balanced placebo design. Active treatment (lidocaine/prilocaine) was applied in the open treatment (T+E+) and hidden treatment (T+E-) conditions. Treatment expectancy was induced in the open treatment (T+E+) and placebo (T-E+) conditions. An individually calibrated temperature corresponding to a visual analog scale (VAS) rating of 6 (0–10) was used for all 4 sessions. (B) Trial design. Each trial presented a condition cue, treatment expectancy rating, pain anticipation, pain stimulation, and pain intensity rating. During each of the 4 sessions, 12 trials were performed.

standard gradient echo planar imaging sequence was employed (repetition time 2.58 seconds; echo time 26 milliseconds; flip angle 80°; field of view $220 \times 220 \text{ mm}^2$; GRAPPA Pat Factor 2). Each volume consisted of 42 transversal slices with a voxel size of $2 \times 2 \times 2 \text{ mm}^3$ and 1 mm gap. Volumes were individually tilted by approximately 30° relative to the AC-PC line. This allowed whole-brain acquisitions including the brainstem. The first 4 volumes of each session were discarded to account for T1 saturation effects. High resolution anatomical T1 scans were acquired using an MPAGE sequence with a voxel size of $1 \times 1 \times 1 \text{ mm}^3$.

2.4. Behavioral data analysis

Behavioral data analysis was performed using SPSS 20 (IBM, Armonk, USA). First, mean pain ratings and mean treatment expectancy ratings were calculated over all 12 trials for each session. Subsequently, a linear mixed effects model using maximum likelihood estimation with 2 repeated-measurement condition predictors (treatment vs no treatment and expectation vs no expectation) was used to analyze the treatment main effect, expectancy main effect, and interaction effects. Because of a time-dependent decrease in the effectiveness of capsaicin during the course of the experiment, we also included the order of the conditions as a covariate in all behavioral analyses of pain rating data. Treatment expectancy ratings were analyzed using a within-subject analysis of variance (ANOVA) with the factors active treatment, expected treatment, and time. Nonsphericity was corrected using the Greenhouse-Geisser method. Results were considered to be significant at $P < 0.05$ (one-tailed, given clear a priori hypotheses).

2.5. fMRI data analysis

fMRI data preprocessing and statistical analyses were performed using SPM8 (Wellcome Department of Imaging Neuroscience, London, UK). Data preprocessing consisted of motion correction (realignment and unwarp), coregistration of the T1 anatomical scan to the functional images, spatial normalization using the transformation parameters obtained from segmentation of the anatomical T1 image and smoothing using a 6 mm (FWHM) isotropic Gaussian kernel. A high-pass filter with a cutoff period of

128 seconds was employed and a correction for temporal autocorrelations was performed using a first-order autoregressive model.

fMRI data analysis was based on a general linear model approach as implemented in SPM8. The first-level analysis included 5 experimental regressors for each session: condition cue at trial start; treatment expectancy rating; pain anticipation; pain stimulation; and pain intensity rating. Additionally, 4 session constants were used, resulting in a total of 24 regressors for every subject.

The onsets of the pain regressors were shifted by 3 seconds relative to the start signal sent to the thermode. This shift accounts for the delay of subjective pain perception relative to the start of the temperature rise [43]. Each regressor was modeled by using boxcar functions convolved with the hemodynamic response function. After model estimation, main effect *t* statistics for each voxel were calculated for each regressor during anticipation and pain stimulation. The ensuing contrast images were then used for a second-level group analysis. One-way within-subject ANOVA was employed, with subjects as a random effects factor and condition as a fixed effects factor. Additionally, as in the behavioral data analysis, the condition order was included as covariate to reduce time-dependent error variance.

For all imaging data analyses, results were considered significant at $P < 0.05$, corrected for multiple comparisons using a family-wise error rate (FWE) approach. In a priori regions of interest, correction was based on a small-volume approach based on peak coordinates obtained from previous studies on pain processing [9,18,46]. For all other brain regions, whole-brain FWE correction was applied. For illustration purposes, statistical maps were thresholded at $P < 0.005$ uncorrected with a voxel extent of minimum 10 and overlaid on the mean structural image of all subjects. All activations are reported using *x*, *y*, *z* coordinates in Montreal Neurological Institute standard space.

3. Results

3.1. Behavioral data

Before entering the scanner, participants expected a significantly lower pain level during trials announced as treatment [open treatment (T+E+) and placebo (T-E+)] compared with the trials announced as control [hidden treatment (T+E-) and control

(T–E–), 3.7 ± 1.8 vs 6.9 ± 1.9 (mean $\pm$ SD), $F(1,31) = 131.0$, $P < 0.001$, indicating that the initial expectancy manipulation was successful.

The average temperature corresponding to a VAS rating of 6 was $39.8 \pm 2.9^\circ\text{C}$ on capsaicin-pretreated skin. Over all experimental sessions, subjects rated the pain intensity with a VAS rating of 5.5 ± 2.9 , indicating that the individual adjustment was successful.

First, we tested for a main effect of lidocaine/prilocaine treatment. Pain ratings showed a significant main effect of treatment (Fig. 2A; Table 1). In the 2 active treatment sessions (open treatment and hidden treatment), subjects exhibited significantly less pain compared with the control cream sessions [placebo and control, $F(1,96) = 44.2$, $P < 0.001$; 4.5 ± 2.8 vs 6.5 ± 2.3], indicating that our lidocaine/prilocaine treatment was successful.

We then compared the 2 expectancy conditions (open treatment and placebo) with the no-expectancy conditions (hidden treatment and control). We did not observe a significant main effect of expectancy on reported pain (5.3 ± 2.8 vs 5.7 ± 2.7). However, the expectancy effect in the treatment conditions (open treatment vs hidden treatment) was significant, ie, volunteers reported significantly less pain in the open treatment condition compared with the hidden treatment condition [$F(1,31) = 3.0$, $P < 0.05$; 4.1 ± 2.7 vs 5.0 ± 2.9].

Finally, we investigated whether our lidocaine/prilocaine treatment interacted with the expectancy manipulation. The expectancy effect was significantly larger for the treatment conditions compared with the no-treatment conditions [(open treatment – hidden treatment) > (placebo – control), $F(1,96) = 2.9$, $P < 0.05$]. This effect was mainly driven by the significant difference between open treatment and hidden treatment (see earlier), while the difference between placebo and control was not significant (6.6 ± 2.3 vs 6.4 ± 2.3).

Treatment expectancy ratings (Fig. 2B; Table 1) were significantly higher for the treatment conditions compared with the no-treatment conditions [$F(1,31) = 13.3$, $P < 0.001$]. Further, treatment expectancy ratings were significantly higher in the expectancy conditions compared with the no-expectancy conditions [$F(1,31) = 20.5$, $P < 0.001$]. There was also a significant 3-way active treatment $\times$ expected treatment $\times$ time interaction [$F(1,31) = 3.9$, $P < 0.05$]. To further investigate, we performed subsequent analysis with one-way ANOVA. We found a significant increase of treatment expectancy over time in the open treatment condition

[$F(1,31) = 7.7$, $P < 0.001$] and a significant decrease over time in the placebo condition [$F(1,31) = 4.9$, $P < 0.05$].

3.2. Imaging data

3.2.1. BOLD responses to painful thermal stimulation on capsaicin-sensitized skin

Phasic pain stimuli (all pain sessions vs baseline) increased activation in a network of regions previously implicated in pain processing [2]. We found bilateral activation in the thalamus [$16, 6, 12$], $t(92) = 14.04$, $P < 0.001$; [$-14, 10, 0$], $t(92) = 11.20$, $P < 0.001$, the primary somatosensory cortex [$62, -38, 40$], $t(92) = 19.26$, $P < 0.001$; [$-58, -42, 46$], $t(92) = 9.41$, $P < 0.001$, the secondary somatosensory cortex [$60, -16, 24$], $t(92) = 18.09$, $P < 0.001$; [$-58, -26, 20$], $t(92) = 18.45$, $P < 0.001$, the insular cortex [$52, 16, -2$], $t(92) = 24.02$, $P < 0.001$; [$-38, 12, -2$], $t(92) = 19.85$, $P < 0.001$, the anterior cingulate cortex [$0, 26, 40$], $t(92) = 15.83$, $P < 0.001$, the dorsolateral prefrontal cortex [$42, 36, 30$], $t(92) = 15.86$, $P < 0.001$; [$-44, 30, 30$], $t(92) = 7.58$, $P < 0.001$ and the cerebellum [$-22, -66, -48$], $t(92) = 12.34$, $P < 0.001$; [$36, -66, -26$], $t(92) = 8.49$, $P < 0.001$.

3.2.2. Neuronal representation of lidocaine/prilocaine treatment effect

We tested whether active treatment would elicit a lower activation in pain-responsive brain regions [(placebo + control) > (open treatment + hidden treatment)]. We observed signal changes in the bilateral anterior insular cortex [$38, 12, 14$, $t(92) = 4.02$, $P < 0.01$; $-38, 4, 14$, $t(92) = 3.95$, $P < 0.05$] in the active treatment conditions compared with the no-treatment conditions (Fig. 3A, in green). The reverse contrast [(open treatment + hidden treatment) > (placebo + control)] revealed no significant results.

3.2.3. Brain regions exhibiting expectation effects in the active treatment conditions

We further tested whether we could observe expectancy-related brain responses in the active treatment conditions (hidden treatment > open treatment). We observed higher signals in the hidden treatment conditions in the bilateral secondary somatosensory cortex [$54, -24, 34$, $t(92) = 3.64$, $P < 0.05$; $-54, -26, 34$, $t(92) = 3.48$, $P < 0.05$], bilateral anterior insular cortex [$-34, 10, 8$, $t(92) = 3.90$, $P < 0.05$; $34, 14, 8$, $t(92) = 3.68$, $P < 0.05$], mid cingulate cortex [$4, 14, 32$, $t(92) = 3.99$, $P < 0.01$] and bilateral thalamus

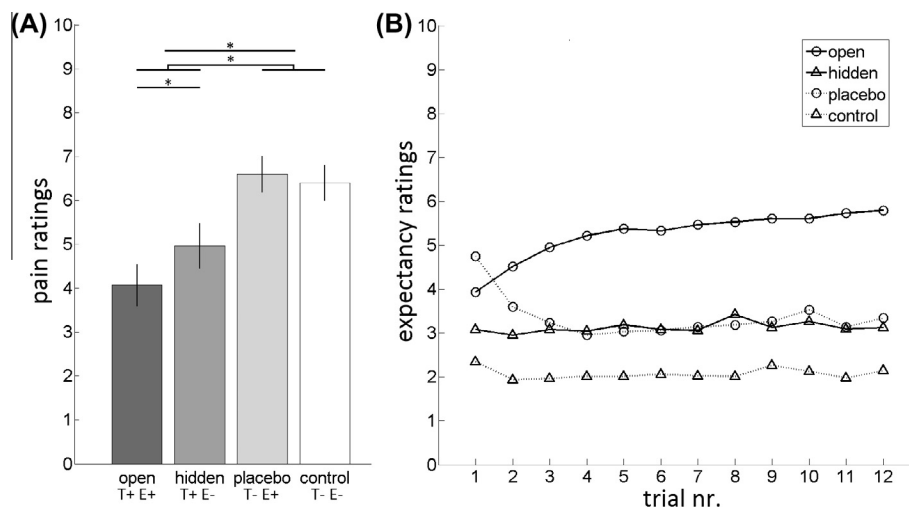


Fig. 2. (A) Mean $\pm$ SE reported pain ratings (visual analog scale) for open treatment (T+E+), hidden treatment (T+E–), placebo (T–E+), and control (T–E–) conditions. We observed a main effect of active treatment, an expectancy effect in the treatment conditions, and an interaction between treatment expectancy and active treatment. (B) Mean expectancy ratings over the course of the sessions. We observed a main effect of active treatment, a main effect of expectancy, a significant increase in expectancy ratings over the trials in the open treatment condition, and a significant decrease in expectancy ratings over the trials in the placebo condition.

Table 1Mean $\pm$ SD treatment expectancy and pain ratings (visual analog scale) for all four conditions.

	Open (T+E+)	Hidden (T+E-)	Placebo (T-E+)	Control (T-E-)
Expectancy	5.26 $\pm$ 2.6	3.12 $\pm$ 2.8	3.35 $\pm$ 2.4	2.08 $\pm$ 2.5
Pain	4.07 $\pm$ 2.7	4.96 $\pm$ 2.9	6.60 $\pm$ 2.3	6.40 $\pm$ 2.3

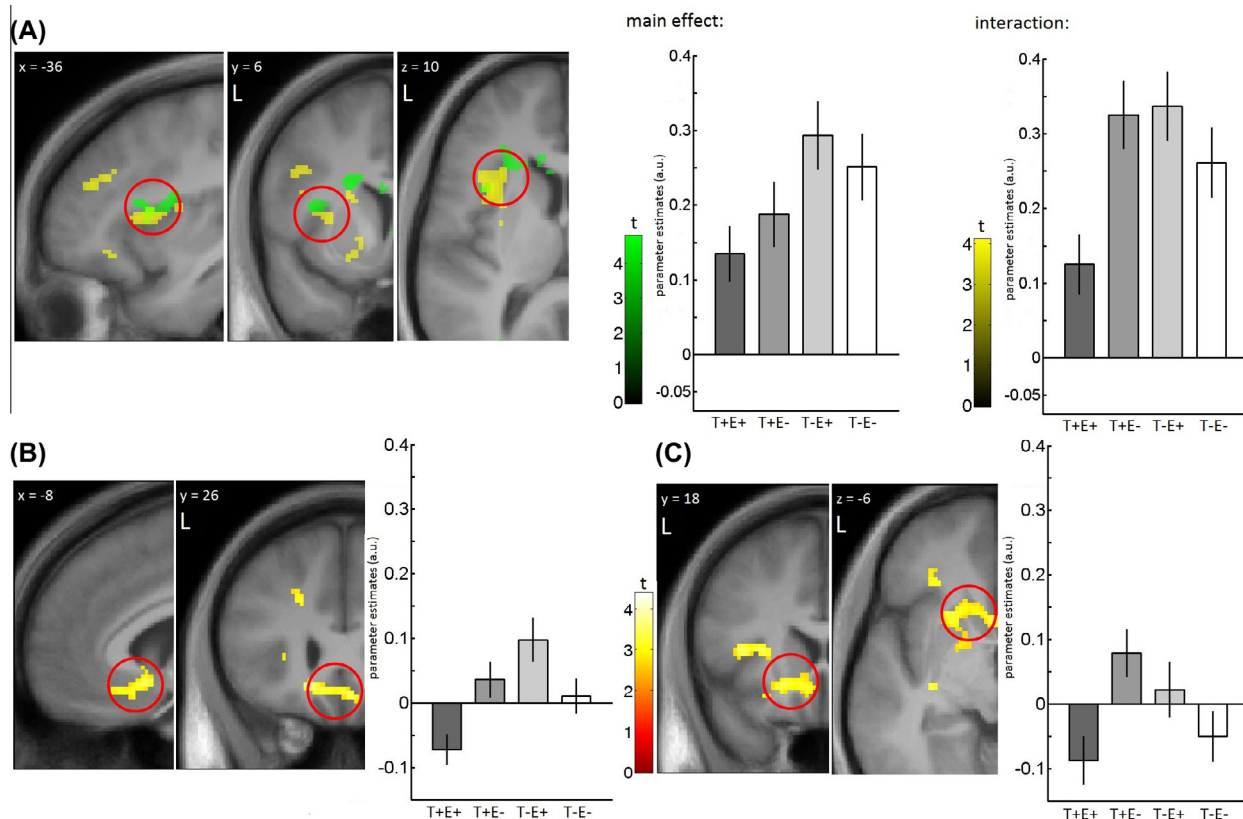


Fig. 3. (A) BOLD activation and parameter estimates $\pm$ SE of the peak voxels associated with the main effect of treatment (green) in the insular cortex ($-38/4/14$). BOLD activation and parameter estimates $\pm$ SE associated with the interaction between treatment expectancy and active treatment (yellow) in the insular cortex ($-34/12/10$), (B) the anterior cingulate cortex ($-16/26/-10$) and (C) the ventral striatum ($-8/16/-10$). The color bar indicates t values and the visualization threshold is set to $P < 0.005$ with a voxel extend of minimum 10.

[12, $-10/2$, $t(92) = 3.66$, $P < 0.05$; -12 , $-2/4$, $t(92) = 3.68$, $P < 0.05$]. This indicates that pain processing areas were more active in the hidden treatment condition compared with the open treatment condition. The reverse contrast (open treatment $>$ hidden treatment) revealed no significant results.

3.2.4. Brain regions exhibiting treatment-by-expectancy interaction effects

Finally, we investigated BOLD signal changes associated with a treatment-by-expectancy interaction effect analogous to our behavioral data. The contrast [(open treatment $<$ hidden treatment) $>$ (placebo $<$ control)] showed a significant activation in the left anterior insular cortex [$-34/12/10$, $t(92) = 3.70$, $P < 0.05$; Fig. 3A, in yellow]. The right anterior insular cortex showed a similar effect but failed to reach statistical significance [$34/14/8$, $t(92) = 2.83$, $P = 0.003$ (uncorrected)]. We also observed a significant effect in the left subgenual anterior cingulate cortex [$-16/26/-10$, $t(92) = 3.76$, $P < 0.05$; Fig. 3B], the left ventral striatum [$-8/16/-10$, $t(92) = 3.79$, $P < 0.05$; Fig. 3C] and bilateral secondary somatosensory cortex [$-58/-26/32$, $t(92) = 3.74$, $P < 0.05$; $56/-26/30$, $t(92) = 3.64$, $P < 0.05$].

Investigation of the parameter estimates underlying this interaction revealed that the insular cortex [$t(92) = 13.28$, $P < 0.001$]

and the secondary somatosensory cortex [L: $t(92) = 6.80$, $P < 0.001$; R: $t(92) = 13.56$, $P < 0.001$] were generally more activated by pain compared with baseline, with a peak in activation during hidden treatment and placebo conditions, while the anterior cingulate cortex and the ventral striatum showed no significant activation during pain compared with baseline. The reverse contrast [(hidden treatment $<$ open treatment) $>$ (control $<$ placebo)] revealed no significant results.

4. Discussion

In this fully crossed study, we observed a significant reduction in reported pain during active treatment and an augmentation of this treatment effect by expectation. Reduced pain ratings during active treatment were accompanied by reduced activity in the bilateral anterior insular cortex. In addition, we observed an interaction between treatment and expectation in the anterior insular cortex, the subgenual anterior cingulate cortex, and the ventral striatum.

This balanced 2 by 2 factorial placebo design allowed us to separate drug-related and expectancy-related effects [42] at the behavioral and neuronal levels. To further increase the clinical validity of our study, we used an established clinical pain model

(capsaicin-pretreated skin). Topical capsaicin selectively binds to the TRPV1 receptor and sensitizes the A δ and C pain- and heat-sensing fibers [12,55]. This method has previously been successfully used as a clinical model for neuropathic pain [36,50] and allowed us to use lidocaine/prilocaine application as an effective treatment [30].

We observed a significant decrease in reported pain during lidocaine/prilocaine treatment and a decrease of the BOLD signal in the anterior insular cortex. The insular cortex is consistently activated during pain [2] and has been linked to thermal pain intensity [38]. As our lidocaine/prilocaine treatment led to a partial block of A δ and C fibers [30], the reduction in insular activity is likely to be due to a reduction in nociceptive input reaching the brain. Importantly, our observation of a reduced BOLD signal in the anterior part of the insular cortex is in agreement with results in clinical pain (allodynia) showing involvement of the anterior insular cortex [45] and with previous results from studies applying heat pain on capsaicin-pretreated skin [33].

The bilateral thalamus, bilateral secondary somatosensory cortex, the bilateral anterior insular cortex and the mid cingulate cortex showed reduced activity in the open treatment compared with the hidden treatment. These brain regions are commonly activated during painful stimulation and show reduced activity if placebo effects are induced [2,56]. Thus, the reduction in activation observed in these regions may reflect a mechanism similar to placebo analgesia.

The interaction between treatment and treatment expectation was expressed in the anterior insular cortex, subgenual anterior cingulate cortex, and in the ventral striatum. The insular cortex has been shown to be consistently involved in pain modulation [53], and activation in the insular cortex has been shown to be modulated by topical lidocaine treatment [23], suggesting the insular cortex as a likely region for an interaction effect of treatment and expectancy. The anterior insular cortex has also been associated with processes other than pain, such as the processing of interoceptive information and subjective emotions [14,15], which are likely to be involved in treatment expectation.

In the subgenual anterior cingulate cortex, we observed a lower activation in the placebo condition compared with the control condition. A similar reduction has been observed in several previous placebo studies [18,56] and this has been linked to antinociceptive processes [17]. Importantly however, in this fully crossed design, we observed that this is accompanied by a higher activation in the hidden treatment condition compared with the open treatment condition.

The anterior cingulate cortex plays a pivotal role in placebo analgesia and is a core region of the descending pain modulatory network [53]. It has consistently been demonstrated that placebo analgesia is associated with increased functional coupling between the anterior cingulate cortex and the periaqueductal gray [8,18,37]. However, the anterior cingulate cortex is also implicated in affective responses to errors [52] and error detection [10,11]. The latter observation is interesting, as the subgenual anterior cingulate cortex showed the highest response in the hidden (T+E–) and placebo (T–E+) conditions (Fig. 3B). These conditions could be labeled “error” conditions as the outcome is in disagreement with the expectation (ie, treatment without expectation in the hidden condition, and expectation but no treatment in the placebo condition). It is thus possible that the subgenual anterior cingulate cortex also signals an error inherent in placebo manipulations and that its role in pain and expectancy processing is more complex than previously assumed [18,56].

The interaction between treatment and expectation was also evident in the ventral striatum. The ventral striatum shows opioid- and dopamine-related activation during placebo analgesia [48,57]. Dopamine release in the ventral striatum has been associated with

the expectation of reward [44]. During placebo analgesia, ventral striatum activity and dopamine release in the ventral striatum are associated with anticipated placebo analgesic response in healthy controls [24,47,48] and Parkinson's disease [22], as well as mismatch between expected and received treatment [47]. In analogy to error processing in the anterior cingulate cortex, the ventral striatum has been associated with prediction error processing during reward learning [25]. Again, it is interesting to note that the ventral striatum shows the highest level of activation for those conditions that generate an unsigned prediction error (ie, more or less treatment than expected).

There is extensive literature on the neuronal correlates of placebo analgesia [2,3,7,28,53], involving the observed brain regions in opioid-related placebo analgesia. Two previous studies investigated the relationship between drug effect and treatment expectancy with a systemic application of the opioid-receptor agonist remifentanyl [4,9]. By contrast, we employed a topical anesthetic treatment, which makes it unlikely that the observed interaction between expectation and treatment is mediated by central opioid effects. In our study, expectation only had an effect in combination with treatment. We therefore propose that expectation had an effect only when pain relief was experienced (through treatment). This is important as it speaks to the fact that an effective treatment can benefit more from a positive expectation than an inert treatment, as is the case in placebo treatments. As the latter is ethically questionable in a clinical context, our data clearly show that it is also less effective than exploiting the expectation effect in the context of an effective treatment.

Atlas and colleagues [4] observed an additive effect of expectation and treatment but no interaction. This might be related to an additive effect of endogenous opioids thought to be released by expectation [18,31] and opiate effects within the central nervous system. In addition, differences from the study by Atlas and colleagues [4] might also be related to the large average treatment effect of 30.5% in our study, compared with 7.8% in the behavioral part of their study. The stronger treatment effect in our experiment could have substantially increased the previously mentioned cognitive effect in our study. A second difference is the employed pain model. While we employed a capsaicin heat pain model, inducing a hyperalgesic condition similar to certain clinical states [36,50], Atlas and colleagues [4] used a standard heat pain model.

This study also has limitations. Although we observed a significant difference in pain expectancy between expectancy and no-expectancy conditions, during the test sessions we did not find a significant main effect of expectation on pain, which is related to the absence of an expectation effect in the no-treatment conditions. This result is likely owing to 2 reasons. First, we did not use a conditioning procedure before the experiment to increase the placebo response during the experiment, as in previous experiments [18,35,41]. Second, the previously mentioned reinforcing effect of the active treatment could have acted in reverse in the placebo condition, when active treatment was expected but no treatment received, thereby reducing the initial treatment expectation even further.

In our experiment, we tested conditions in different sessions. This allowed us to reposition the thermode between sessions. It also minimized the confusion concerning which condition was being tested for subjects. While this method increases the possibility for order effects, these effects were controlled for in our experiment.

In conclusion, even though our treatment was topical, our results demonstrate that interactions between treatment and expectation occur. Active treatment had a significant effect on reported pain, but treatment in combination with treatment expectancy yielded an additional significant reduction in reported pain. This finding is very relevant to clinical practice and points to the

important effect of treatment-related positive expectancies substantially enhancing therapeutic outcomes. Therefore, the deliberate employment of expectancy strengthening strategies in clinical practice offers an important opportunity to increase the therapeutic benefit for the patient.

Conceptually, the observed interaction effect is also relevant for the design of clinical trials, in which additivity is assumed to separate treatment effects from expectancy effects. Although the observed interaction in our study did not obscure the effectiveness of the treatment, it is possible that expectation would not boost a treatment effect, as shown here, but that a treatment effect would only be detectable in the context of expectation. This has previously been demonstrated in the case of open vs hidden treatment with metamizole (dipyrone) [13], which had almost no effect when given without expectation.

Conflict of interest statement

The authors do not have any conflicts of interest.

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